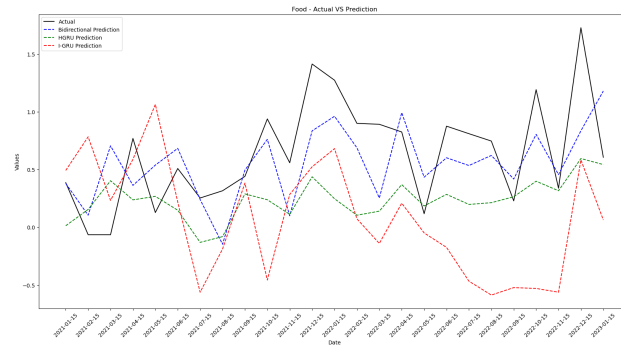


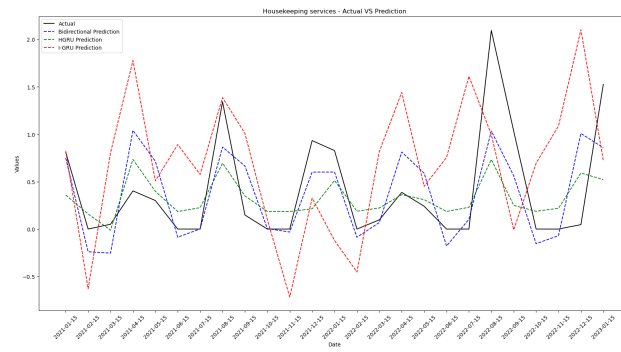
models: BiHRNN, HRNN, and I-GRU in blue, green, and red, respectively. As shown in the graphs, the BiHRNN model demonstrates superior predictive accuracy, achieving lower RMSEs and more effectively capturing shifts in trends compared to its counterparts.

Figure 5.1. Disaggregated Index CPI Predictions

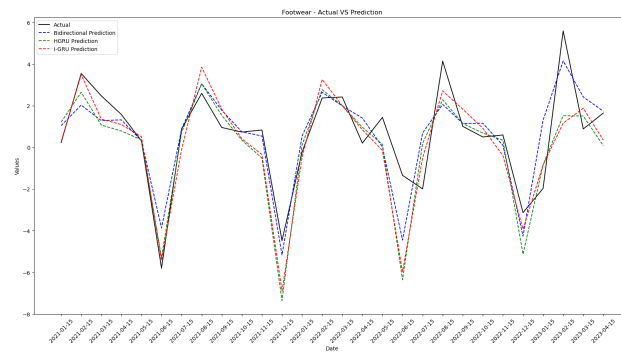
(a) Food



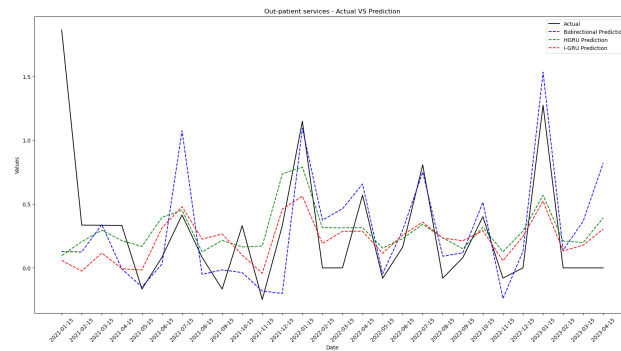
(b) Housekeeping



(c) Footwear



(d) Out-patient Services



6 Conclusion

This thesis presented a novel approach to inflation time series forecasting using Bidirectional Hierarchical Recurrent Neural Networks (BiHRNNs). The primary objective was to leverage hierarchical data structures to enhance prediction accuracy at both granular and aggregate levels of Consumer Price Index (CPI) data. The proposed BiHRNN model was applied to CPI data from three established markets: Canada, Norway, and the United States.

The BiHRNN model was trained and evaluated on these datasets, with hyperparameters carefully optimized for each market. Additionally, we experimented with loss functions designed to propagate information between hierarchical levels, which proved crucial for achieving superior forecasting accuracy. Compared to traditional RNNs and machine learning models, the BiHRNN demonstrated significantly improved performance across all hierarchical levels.

Future work will extend the application of BiHRNNs to CPI datasets from other countries and sectors, addressing the unique challenges they present. Further enhancements will focus on incorporating additional features and exploring advanced methods for parameter optimization and loss function design.

In summary, this thesis established BiHRNNs as a robust framework for inflation time series forecasting, highlighting their ability to surpass traditional methods and laying the groundwork for further research and innovation in this area.